1. Water Jug Problem

Report:

The Water Jug problem was successfully solved by performing a series of operations such as filling, emptying, and transferring water between the two jugs. The objective of obtaining exactly 2 gallons of water in the 4-gallon jug was achieved. This experiment illustrates the concept of state-space representation and search techniques used in Artificial Intelligence for solving real-world problems.

2. Mars Rover Problem

Report:

The Mars Rover problem was analyzed by identifying its percepts, environment, actions, performance measures, and suitable agent architecture. It was concluded that a Utility-Based Agent is the most appropriate model for the Mars Rover as it enables intelligent decision-making under uncertain conditions. This study highlights the importance of intelligent agents in autonomous space exploration.

3. 8-Queens Problem

Report:

The 8-Queens problem was solved by determining a valid arrangement of queens on the chessboard such that no two queens attack each other. The problem demonstrates the use of search algorithms and constraint satisfaction techniques in Artificial Intelligence. The successful placement of all eight queens confirms the effectiveness of the chosen approach in solving complex combinatorial problems.

4. OLA Cab Booking Problem

Report:

The OLA Cab Booking problem was modeled using a Goal-Based Agent, which selects the most suitable cab based on the user's preferences and travel requirements. The experiment demonstrates how intelligent agents can be applied in modern transportation systems to improve user experience, optimize resource allocation, and provide efficient service recommendations.

5. Uniform Cost Search (UCS) Problem

Report:

The Uniform Cost Search algorithm was successfully applied to determine the least-cost path between the source and destination warehouses. The algorithm expanded nodes based on their cumulative costs and guaranteed an optimal solution. This experiment demonstrates the significance of UCS in pathfinding and optimization problems commonly encountered in logistics, navigation, and network routing applications.